

Nanoparticle Treatment for Inflammatory Bowel Disease (IBD) using
PolyLactic-co-Glycolic Acid (PLGA) as Drug Delivery System

Nicole Abel (20890284)

Joan Gabrielle Basa (20881535)

Limia Mohamed (20876811)

Abstract

PLGA, or Poly Lactic-co-Glycolic Acid, are a group of FDA-approved mechanically stable, biodegradable, and biocompatible polylactic acid (PLA) and polyglycolic acid (PGA) copolymers that are used as a method of drug delivery. Throughout the years, their effectiveness in drug delivery is studied to develop new drug release devices or systems. PLGA-based nanotechnology is mostly used in controlled delivery of small molecule drugs, proteins, and other macromolecules such as those involved in treating inflammatory diseases, vaccine administration, and tissue engineering. Inflammatory Bowel Disease, which includes Crohn's disease and ulcerative colitis, occurs due to chronic inflammation in the gastrointestinal tract. Although the cause of these diseases remains unknown, it is thought to be caused by an overactive immune response to certain microbes in the gut. For this reason, there is no cure and the disease requires treatment to reduce inflammation. This literature review will focus on the use of PLGAs loaded with 5-aminosalicylic acid or budesonide to help treat IBD.

Table of Contents

Introduction	4
Background Information	5
Nanotherapy	5
Nanoparticles/Nanocarriers	6
Polymeric Nanoparticles	6
Main Content	8
PLGA as Drug Delivery System	8
Properties and Chemical Characteristics of PLGA	8
Synthesis of PLGA	8
Preparation of PLGA for Nanotherapy	9
Inflammatory Bowel Disease (IBD)	11
Symptoms and Diagnosis, Comparison with Ulcerative Colitis and Crohn's Disease	11
Genetic factors	11
Environment factors	12
Microbial factors	13
Applications of PLGA to IBD Treatment	14
PLGA with 5-aminosalicylic acid (5-ASA)	15
PLGA with Budesonide	16
PLGA with tasquinimod (TAS) and macrophage membranes (MMs)	17
PLGA with SNX10-shRNA plasmids (SRP)	17
PLGA with Garcinol	18
Implications	19
Conclusion	20
References	21

Introduction

Advancements in biological and medical physics brought forth through years of dedicated efforts have greatly contributed to improving human life. One example of such remarkable advancement is the discovery of nanoparticles and their potency as a drug delivery system intended to treat various medical conditions. A nanoparticle, as the name implies, is a particle that is measured on the nanoscale (10^{-9} m) with sizes often ranging from 1 to 100 nanometers in diameter. As drug carriers, they exhibit a promising potential in delivering drug(s) to the target body organ or affected site with little to no side effects [7][26]. Drug-loaded nanoparticles can be delivered to the target site through intravenous, oral, or rectal administration.

In this review, the main objective is to explore and examine the nanoparticle poly lactic-co-glycolic acid (or “PLGA”) as a drug delivery system for treating inflammatory bowel disease (IBD). IBD is a disease characterized by chronic inflammation in the digestive tract and mostly causes abdominal pain, diarrhea, and rectal bleeding as well as other symptoms such as fever and fatigue. The exact cause or underlying mechanism of this disease is unknown, but genetic, environment and microbial factors can all give rise to its development. IBD affects the gastrointestinal system and in turn, can negatively impact the entire body. Studies have presented that PLGA drug delivery demonstrates the ability to provide steady drug release with accumulative tendency, biodegradability and reduced adverse effects contributing to its potential as a drug carrier.

The drugs and compounds of interest for this review are 5-aminosalicylic acid, budesonide, tasquinimod, SNX10-shRNA plasmids and garcinol. 5-aminosalicylic acid (5-ASA) and budesonide are two of the most commonly-used drugs for IBD treatment. 5-ASA is known to inhibit the production of cytokines and inflammatory mediators, while the budesonide’s specific mechanism of action is unknown [21][22]. Looking at ulcerative colitis (UC), a form of IBD, tasquinimod is an inhibitor of the inflammatory protein S100A9 which has elevated levels in UC. SNX10-shRNA plasmids, on the other hand, are proteins that inhibit the progression of UC. Moreover, garcinol is a compound that is a strong anti-inflammatory agent.

The PLGA application to IBD treatment section of this review will highlight the models and materials used, method of preparation, characterization, and the release mechanism of these drugs and compounds.

Background Information

Nanotherapy

Nanotherapy can be described as the use of nanotechnology consisting of nanoparticles and/or nanosize devices for therapy treatments. These nanotechnologies tend to range in size from 5-500nm in at least one dimension [10]. The research done on nanotherapies focuses on the use of nanoparticles to improve treatments such as increasing the drug solubility or concentrating the drug at a particular site [10].

The following figure shows the various types of nanotechnology used and the breakdown of nanoparticles.

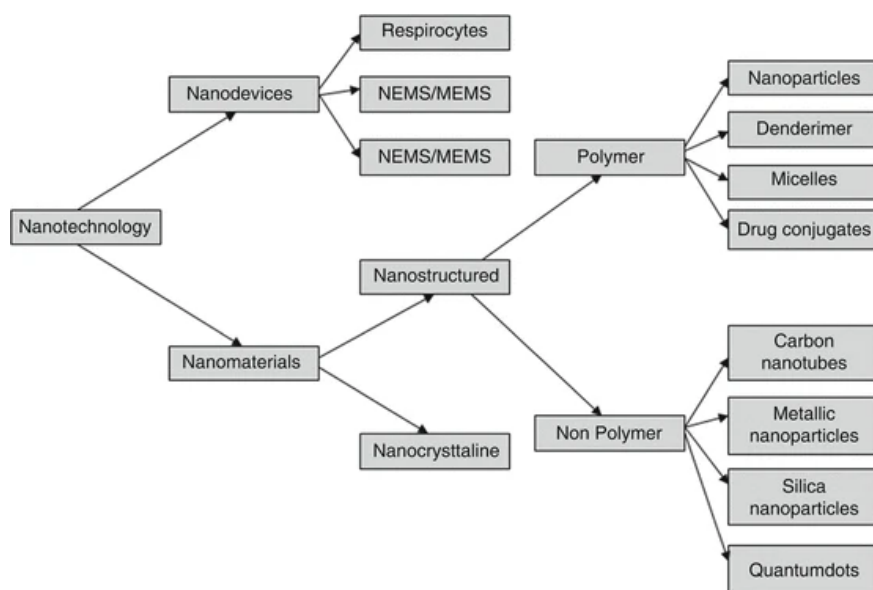


Figure 1. An hierarchy diagram illustrating different pharmaceutical nanosystems. The molecule of interest in this review follows the “nanomaterials - nanostructured - polymer - nanoparticles” pathway [4].

Nanoparticles are one such nanotechnology that is used and has been studied extensively in various treatments ranging from inflammatory diseases to an alternative cancer treatment [10]. Nanoparticles can be administered in many ways for drug delivery depending on the application. Just like other drugs, they can be taken orally such as through solid pills and pulmonary methods, injections and transdermal means [9][19]. Due to the wide variety of delivery means, there are a lot of different applications that nanoparticles can be used for.

Nanoparticles/Nanocarriers

Like the name suggests, nanoparticles are “particles” that are measured on the nano scale and are one size that tends to be between 1-100 nm in diameter [25]. Where the nanoparticle itself is the drug constructed at the nanoscale [25]. Some nanoparticles are larger than 100 nm depending on the application; these particles can be a way of carrying various drugs and act as a “carrier”, also known as nanocarriers [8].

Nanoparticles used for drug delivery systems tend to range in size from 10-1000nm in size since they require an adequate amount of medication within the particle itself [8].

Due to their nanoscale size, nanoparticles may end up being more chemically reactive and their behaviour is unpredictable since they obey the laws of quantum physics [25]. As well, the use of nanoparticles in nanotherapy must also be compatible with certain biological processes to stay intact in vivo (inside the living organism) but then be able to be degraded after the drug has been delivered [9]. Generally, nanoparticles must be minimally toxic, compatible with blood, low immunogenicity and be able to be degraded easily by the body [9]. There are many categories/types of nanocarriers which are based on the type of material used and have different applications [9].

Polymeric Nanoparticles

While there are many different polymer and non-polymer nanoparticle types, the focus for this literature review will be of the polymer type. Polymer nanocarriers and nanoparticles are commonly used for drug delivery as they have many characteristics needed for the application including its biodegradability and ability to control drug release [9]. They also tend to be fairly cheap in manufacturing and can be engineered to have specificity [4]. These particles can further be broken down into either natural or synthetic polymers.

Many of the natural polymers used are derived from polysaccharides and proteins [9]. Of which, some of the most common include gelatin, starch, albumin and chitosan [4]. Synthetic polymers are generally made from polyesters that include polyglycolic acid (PGA), polylactic acid (PLA), copolymer poly(lactic-co-glycolic acid) (PLGA), Poly(methyl) methacrylate (PMMA), and many more [9]. Out of these, both PLGA and PLA have been approved for human use by the U.S. Food and Drug Administration (FDA) [9].

These polymeric nanocarriers are generally prepared in one of two different forms. One being what is called “nanospheres” where the drug is uniformly distributed throughout the nanoparticle resulting in a solid spherical matrix consisting of the drug and the polymer used [9][4]. The other is called “nanocapsules” where the drug is in a cavity that is surrounded by the polymer itself [9][4]. The following is a simple schematic that visually shows the difference between the two types.

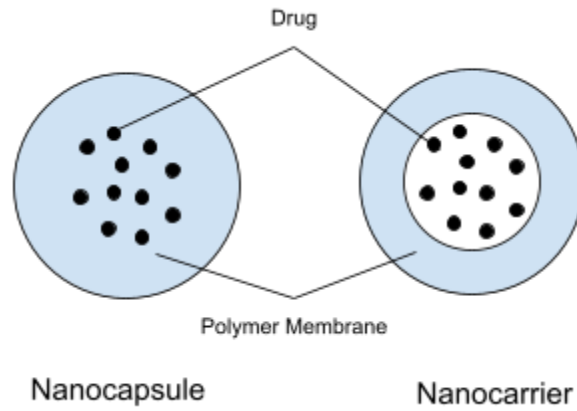


Figure 2. A graphic illustration comparing a nanocapsule vs. a nanocarrier. The nanoparticle PLGA is prepared as a nanocarrier that delivers its load (drug) towards the target site [9][4].

Main Content

PLGA as Drug Delivery System

As previously mentioned, one of the polymeric particles used is poly lactic-co-glycolic acid (PLGA). PLGA is a synthetic copolymer made up of poly lactic acid (PLA) and poly glycolic acid (PGA) [16]. The “co” part of PLGA is in reference to PLGA being made up of a ratio of D- and L- lactic acid forms in comparison to PLDA and PLLA which are made of exclusively the D- or L- lactic acid forms [16]. PLGA is biodegradable and it undergoes hydrolysis by the body going through three main stages; first it is able to be hydrolyzed without an enzyme (stage 1), then it undergoes an enzymatic process to break the components down into lactic and glycolic acid (stage 2) where lactic and glycolic acid are finally metabolized in the Krebs cycle (stage 3) [19]. Due to its biodegradable properties, PLGA is able to be made into different forms and sizes for various applications [16].

Properties and Chemical Characteristics of PLGA

As with other nanoparticles, PLGA is biodegradable and nontoxic [18]. Due to the mixture of PGA and PLA, this allows PLGA to have certain chemical properties compared to each of the components separately and allows for PLGA to be modified. For example, PLGA copolymers with higher lactic acid content, that is PLA, are not as hydrophilic due to PLA having a methyl chain and degrade more slowly whereas it is known that PLGA containing equal parts PGA and PLA will degrade the fastest [18].

The elasticity of the PLGA polymer can be changed with adding more polymeric layers and by increasing the number of cross links between these layers [18]. Along with this structure modification, other polymers can be added to PLGA in order to change its physical and chemical properties. One example is added poly ethylene glycol (PEG) to improve its aqueous solubility [18].

Synthesis of PLGA

Since PLGA is a copolymer, it requires the synthesis from the two polymers PLA and PGA. The main synthesis methods include polycondensation and ring opening polymerization [18]. Polycondensation is the condensation reaction between lactic acid and glycolic acid. Ring opening polymerization is generally used for higher molecular weight PLGA and uses lactide and glycolide [23]. Each of these synthesis methods can be broken up into different subtypes based on different parameters. This allows for slightly different modifications to the overall PLGA product such as its overall molecular weight [18]. The following figure shows the chemical formula of the synthesis for the synthesis of PLGA using polycondensation.

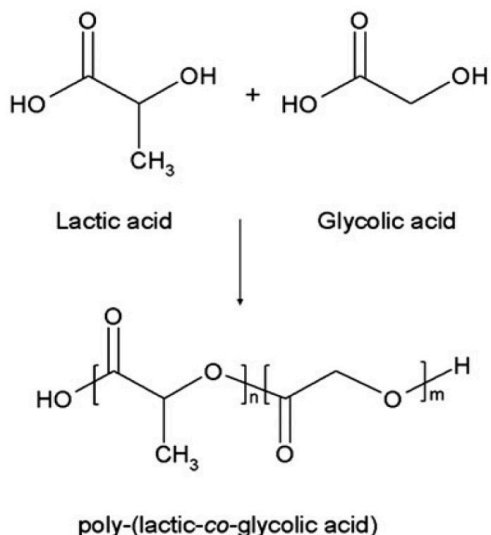


Figure 3: The monomers and chemical structure of PLGA [11].

The following figure shows the synthesis of PLGA using the ring opening polymerization method.

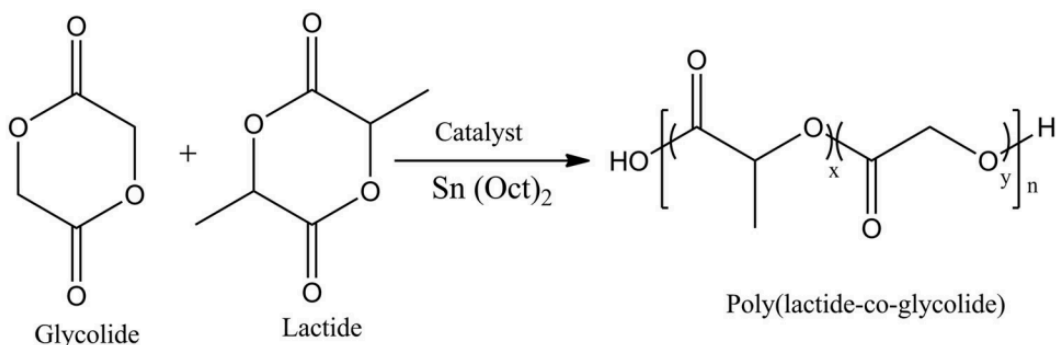


Figure 4: Ring-opening polymerization of PLGA [23].

Preparation of PLGA for Nanotherapy

There are multiple different methods in preparing nanoparticles for nanotherapy. In general though, PLGA is made using different emulsion methods. The two main ones are emulsion solvent evaporation and emulsion solvent diffusion.

The most common methods in making PLGA nanoparticles are the single-emulsion-solvent evaporation and double-emulsion evaporation [15]. In the single-emulsion process, the PLGA is dissolved in a volatile organic solvent that is water-immiscible [15]. The drug of choice is then added to the solution so that there is now a solution/dispersion of the drug particles [15]. This drug-polymer solution is then emulsified in the presence of an emulsifier, usually poly(vinyl alcohol) PVA, where the drug-polymer solution is then stirred in a larger volume of water with an appropriate temperature [15]. This allows for small drug-polymer droplets to form in the large water

bath at which either evaporation or extraction is done so that the oil droplets are retrieved [15]. Due to the drug being added into the polymer-organic solution, it is best for drugs that are water-insoluble so that they can easily dissolve [15]. The double-emulsion process is similar to the single-emulsion process with an extra step added where the drug and polymer are emulsified first before proceeding to the following steps [17]. This extra step is useful for when drugs are water-soluble such as peptides and proteins [15].

Another emulsion technique is the “spherical crystallization technique” which is also known as the emulsion solvent diffusion method [19]. Through this process, PLGA and the drug is first dissolved in an organic, water-miscible solvent and then is added to water where nanoparticles form due to the PLGA’s inability to dissolve in the water and can emulsify [19]. By the end of the process, the drug is now in a PLGA capsule and the prepared PLGA can now be considered a nanocapsule. Using this process has allowed for a wide variety of drugs to be used since different solvent combinations can be used in order for proper nanoparticle formation [19]. This technique is similar to the emulsion solvent evaporation techniques mentioned above. The difference is that in this technique, the polymer precipitates out whereas in the previous technique that does not occur [6].

The final technique that is commonly used is nanoprecipitation and through this technique, the nanoparticles are formed in one step [27]. Through this technique, two miscible solvents are mixed together with the polymer and drug being able to dissolve in only one of the two solvents [27]. For this technique, water is not used as a solvent [5]. This technique can be used for both hydrophobic and hydrophilic drugs [27]. The following simple flow chart explains the main differences between the three major techniques.

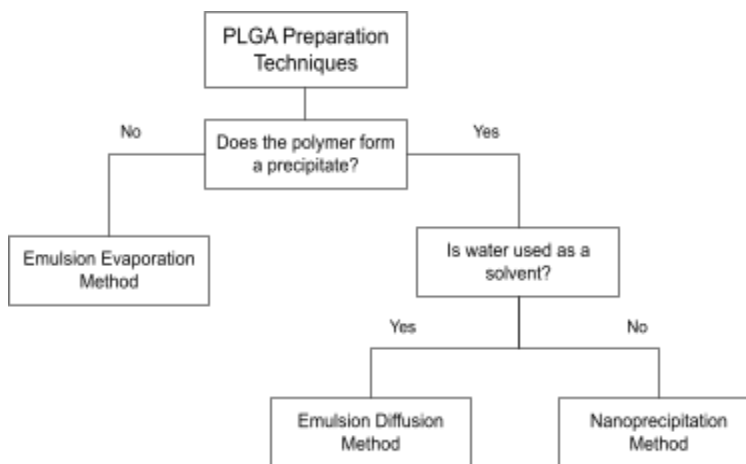


Figure 4. A flowchart showcasing various PLGA nanoparticle preparation techniques. For this review, PLGA is prepared by emulsion evaporation method.

Inflammatory Bowel Disease (IBD)

Symptoms and Diagnosis, Comparison with Ulcerative Colitis and Crohn's Disease

Inflammatory Bowel Disease, which includes both ulcerative colitis and Crohn's disease, is chronic inflammation in the gastrointestinal tract [20]. Ulcerative colitis is characterized by mucosal inflammation in the colon [24]. Meanwhile, Crohn's disease is characterized by transmural inflammation anywhere in the digestive tract [24]. Mucosal inflammation will preserve the five layers of the intestinal wall, while transmural inflammation results in interruptions and loss in the five layers of the intestinal wall [24].

Some common symptoms found in both ulcerative colitis and Crohn's disease are abdominal pain, diarrhea, rectal bleeding which are a result of the inflammation in the digestive tract [20]. In addition there will be secondary symptoms which include weight loss, fever and fatigue; these are a result of the previously mentioned symptoms. Furthermore, ulcerative colitis patients will often experience bloody stools [24]. However, this symptom is uncommon in Crohn's disease patients [24].

Once such symptoms have been observed in patients, inflammatory bowel disease will be diagnosed by upper gastrointestinal radiography, endoscopy, colonoscopy, and gastrointestinal tract sampling methods [24]. These methods will generally allow for the direct visualization of the inflamed gut [20].

Although the cause of the disease remains unknown, the pathogenesis of inflammatory bowel disease has been widely studied. The nature of pathogenesis of IBD can be examined by looking at genetic, environment and microbial factors [1].

Genetic factors

There are many factors which may increase genetic susceptibility of inflammatory bowel disease. This is due to mutations that may be found in genes involving these levels: innate immunity, antigen presentation molecules, epithelial integrity, drug transporter and cell adhesion [28].

Firstly, in the innate immune system we will have molecules including toll-like receptors and nucleotide oligomerization domains that contribute greatly to mucosal homeostasis in the digestive tract [28]. Any mutations or alterations in the following molecule may have an affect on the pathogenesis of inflammatory bowel disease. In particular, there are studies that have looked at specific mutations in NOD2, a cytoplasmic protein, that has roles in recognizing a ligand that fragments of peptidoglycan present in bacterial cell walls [28]. This ligand will induce proinflammatory cytokines and has been associated with increased susceptibility of Crohn's disease [28]. In addition, NOD1, a protein that participates in colonic epithelial defense against particular bacteria (i.e. *E. Coli* and *S. flexneri*), have been studied [28]. Mutations to NOD1 are also shown to be

associated with the development of inflammatory bowel disease[28]. Alterations in toll-like receptors which will be found on the surface of immune and epithelial cells are also said to be associated with inflammatory bowel disease as well [28].

Secondly, when looking at antigen presentation molecules, particularly human leukocyte antigen (HLA) class II molecules, they have a crucial role in immune responses [28]. Particular HLA gene alleles are associated with ulcerative colitis[28]. Furthermore, HLA class III molecules that code for tumor necrosis factor alpha are also associated with inflammatory bowel disease, since tumor necrosis factor alpha will influence cytokine production [28].

Thirdly, there are organic cation transporter proteins (OTCN), involved in epithelial integrity [28]. They will transport a particular compound which is involved with transporting fatty acid [28]. Mutations in OTCN1 AND OTCN2 are associated with the development of ulcers [28]. This is because the inhibition of fatty acid oxidation in the epithelium of the colon has been shown to be associated with the development of ulcerative colitis [28]. Furthermore, mutations in DLG5 genes which are responsible for intestinal epithelial cells will be involved in the increase of intestinal permeability [28].

Fourthly, when looking at multidrug-resistance (MDR-1) genes. These were shown to be significantly high inflammatory bowel disease patients who required surgery after failed medical treatment [28].

Lastly, there are particular cell adhesion alleles that are associated with inflammatory bowel disease. These molecules will have a role in coordinating the “homing of activated inflammatory cells into the gut” [28].

Environment factors

There are many environmental factors that can be associated with the prognosis and progression of inflammatory disease. These factors include: diet, food additives, smoking, drugs, infection, geography, air and water pollution [1]. Smoking has been shown to have protective effects on the development and progression of ulcerative colitis, with lower relapse than non-smokers [1]. The opposite is seen in patients with Crohn's, smoking has negative effects on the development and progression of Crohn's disease [1]. To add, many studies have found that patients with inflammatory bowel disease will have vitamin D deficiency and that low levels of vitamin D is associated with an increased risk of inflammatory bowel disease [1]. There is no evidence for stress causing inflammatory bowel disease. But there are studies which have shown that stress is associated with increased relapse in inflammatory bowel disease [1].

Microbial factors

When comparing the gut microbiota of healthy individuals and disease individuals, there is a notable difference [29]. There is reduced biodiversity in the fecal microbiome of patients with inflammation compared to healthy individuals [29]. In the case of healthy individuals, they will usually have a gut microbiota rich in Firmicutes and Bacteroidetes bacteria [29]. These bacterias will have a role in the production of epithelial metabolic substrates [29]. However, the gut microbiota of patients with Crohn's will usually lack these bacterias and rather be rich in enterobacteria [29]. Meanwhile, ulcerative colitis patients seem to have less *Clostridium* spp and more *E.coli* present in their gut microbiota [29].

Applications of PLGA to IBD Treatment

As a complex organ, the gastrointestinal (GI) tract consists of many different parts and each part works together to perform a wide range of functions such as ingestion and digestion, absorption, and excretion. Defects and diseases in the gastrointestinal tract can negatively impact the entire bodily system. Given this context, nanoparticle-based drug delivery systems are considered as one of the most promising methods of treating IBD in a sense that it has the ability to provide steady drug release while also lessening adverse effects. Earlier, it was already mentioned that some of the characteristics of PLGA nanoparticles which demonstrate its effectiveness as a nanocarrier include: high accumulative properties around the target site, biodegradability, and minimal side effects [7].

Since the discovery of its drug carrying capabilities, numerous anti-inflammatory drugs and compounds have been studied with PLGA nanoparticles to determine the most effective route of administration—intravenous, oral, or rectal. PLGA nanoparticles were loaded with anti-inflammatory drugs such as budesonide, 5-aminosalicylic acid (5-ASA) and tasquinimod, and compounds such as SNX10-shRNA plasmids and garcinol. These anti-inflammatory drugs and compounds help to alleviate symptoms of IBD. Their curative ability is dependent on the severity of the disease [7]. This section will elaborate on the applications of PLGA to IBD treatment, specifically PLGAs loaded with some commonly-used anti-inflammatory drugs and compounds.

Most of the PLGA delivery system designs involve the following methods: preparation of PLGA, culturing of stem cells, characterization of nanoparticles, microscopic imaging, determination of drug content and percent yield, in-vitro drug release, and localization.

For most of these studies, they use the two following formulas in order to determine the amount of drug loaded into the capsule and how effectively capsulated it was.

The percent encapsulation efficiency is calculated by the formula [3][12]:

$$EE\% = \left(1 - \frac{C_f}{C_t}\right) \times 100\%$$

where EE is the encapsulation efficiency, C_f is the free drug concentration and C_t is the total concentration [3].

The percent drug loading is calculated by the formula [3]:

$$LE\% = \frac{W_e}{W_m} \times 100\%$$

where LE is the percentage of the drug loaded in the nanoparticle, W_e is the weight of the encapsulated drug and W_m is the total weight of the drug loaded nanoparticle [3].

PLGA with 5-aminosalicylic acid (5-ASA)

5-aminosalicylic acid, or 5-ASA, is one of the most common anti-inflammatory drugs used to treat inflammatory bowel diseases. It is known to suppress the production of cytokines and inflammatory mediators, but the mechanism underlying the intestinal effects of 5-ASA remains unknown [22]. In Davoudi's (2017) experiment, intestinal organoids were used as a model for the human intestine. Intestinal organoids, which are three-dimensional or 3D in vitro models of the human intestinal epithelium, are used in research for the development and properties of the intestine. They have the ability to absorb small nanoparticles towards the intestinal cavity and attach to the target site [7].

PLGA nanoparticles are loaded with 5-ASA (drug) or Rhodamine B (dye) in four different drug-to-dye weight-by-weight (w/w) ratios. The method of preparation used was double-emulsion (water/oil/water)-based solvent evaporation or extraction [7]. The PLGA/5-ASA and PLGA/Rhodamine B suspensions were then analyzed to evaluate their properties before loading them inside the organoid units [7][12]. The intestinal stem cells (ISCs) were extracted from mice. The mice intestines cut into small pieces were washed with PBS then added with ethylenediaminetetraacetic acid (ECTA), and shaken until the final solution is derived and centrifuged [7]. The extracted cells are then suspended with Matrigel. An experimental control was also prepared which is an organoid culture in the absence of the PLGA nanoparticle. Inverted microscopy was used to track the rate of organoid growth while confocal fluorescent microscopy was used to detect the presence of nanoparticles inside the intestine cavity (lumen) [7].

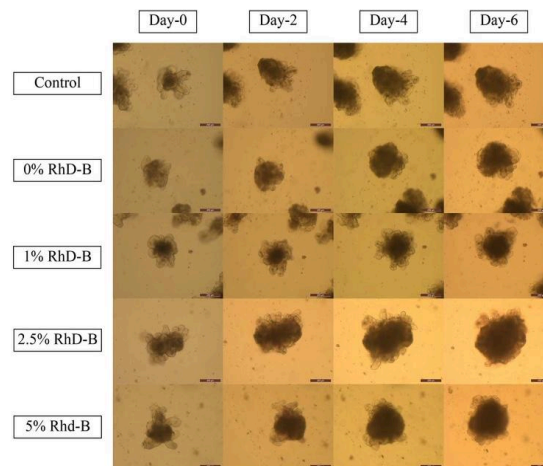


Figure 5. Above shown is the 10x magnification of organoid growth along a 6-day time scale viewed through an inverted microscope. Scale bars used represent a length of 200 μm . All samples showed a progressive growth rate after 6 days, indicating the drug-loaded nanoparticle has no effect on organoid growth [7].

Davoudi et al. (2017) found that the drug-loaded nanoparticles did not affect the growth and viability of the organoid. With the PLGA nanoparticles having a negative zeta potential, they may easily attach to the positively-charged inflamed region and release the 5-ASA anti-inflammatory drug in the area.

PLGA with Budesonide

Budesonide, a steroid or cortisol-like medication, is the drug of choice for treating inflammatory disorders like IBD. Although budesonide has long proven its ability to reduce inflammation and other symptoms of the disease such as diarrhea and cramping, it has exhibited strong adverse reactions. Such side effects can be lessened through the delivery of drug towards the target site [Ali et al., 2016]. In other words, for a more effective approach to treating IBD, a delivery system that is able to carry the drug completely to the target site is the most preferable option. This is where nanoparticles prove their role as a method of drug delivery [2][13].

In an experiment conducted by Ali et al. (2016), PLGA nanoparticles were loaded with budesonide. The method of preparation used was also solvent evaporation. Weighted budesonide was dissolved in an ethyl acetate solvent with PLGA. The encapsulation efficiency, or EE, of budesonide-loaded PLGA nanoparticles was determined through both direct (from measuring the actual mass of the compound) and indirect (measuring the mass for the total drug minus the mass of the free drug) methods [2]. The encapsulation efficiency was about $85 \pm 3.5\%$, quantified by both direct and indirect methods with a percent yield of 94%.

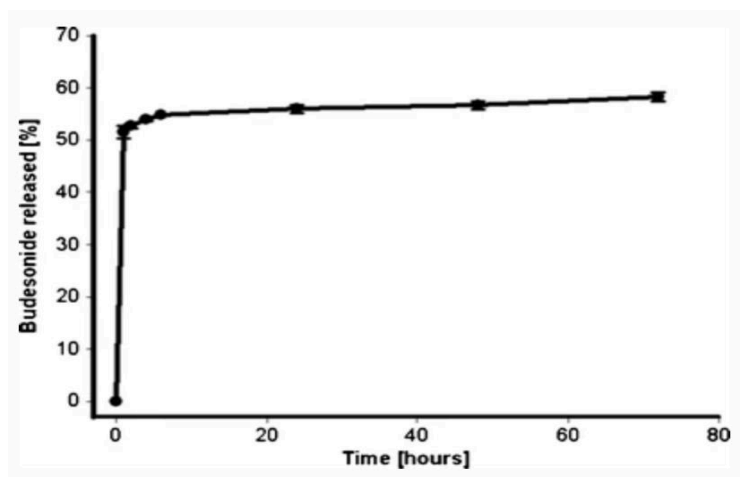


Figure 6. Plot describing the percentage of budesonide released over time in hours. The percent of budesonide released had an initial burst and then achieves a steadier state in over 20 hours [2].

Both healthy and inflamed murine colon tissues received the same dosage of budesonide-loaded PLGAs by oral administration [2]. The abdomens were then analyzed for results. It was shown that for the healthy group, the nanoparticles spread in the whole gut while for the inflamed group the nanoparticles were more accumulated at the target site [2].

PLGA with tasquinimod (TAS) and macrophage membranes (MMs)

Li et al. (2022) designed an orally-administered drug delivery system that cures ulcerative colitis in mice models [14]. The design system was found to have greatly increased the accumulation of drug around the inflamed site, reducing potential exposure of the drug to the non-target areas. However, unlike the other systems where the PLGA is loaded with a treatment drug, the design instead utilizes S100A9 inhibitor tasquinimod (TAS) loaded onto PLGA to synthesize PLGA-TAS nanoparticles [14]. This TAS-loaded PLGA was prepared by the solvent evaporation method. S100A9 is an inflammatory mediator that is shown to be in increased levels in ulcerative colitis and corresponds to the severity of the disease. This molecule is the target for ulcerative colitis treatment and this forms the groundwork of designing the targeted delivery system. The cumulative ability of the PLGA-TAS nanoparticle is brought forth in the overexpression macrophage membranes (MMs) by TLR4, a receptor of the S100A9 ligand, fully obtaining the MM-PLGA-TAS nanocarriers [14].

The study found that MM-PLGA-TAS reduced the symptoms of ulcerative colitis. Specifically, MM-PLGA-TAS achieved its modulative and suppressive effects by reducing S100A9 and other cytokines or inflammatory proteins in the colitis region [14]. This proves that the delivery system designed is a new approach to consider in the treatment of ulcerative colitis, an inflammatory bowel disease (IBD).

PLGA with SNX10-shRNA plasmids (SRP)

Sorting nexins (SNX) are phosphoinositide-binding proteins and they can be a part of some physiological process [3]. SNX10 in particular has been associated with a variety of autoimmune diseases including IBD. If SNX10 is not expressed, it has been seen to inhibit the progression of ulcerative colitis (UC). The 2021 study by Bao et al. had prepared and tested oral PLGA nanoparticles consisting of SNX10-shRNA plasmid (SRP) [3]. This was done via the double-emulsion solvent evaporation method. The nanoparticles use the method of using short hairpin RNAs (shRNAs) where they would create an interfering sequence to SNX10. It is thought that the SRP-NPs would be able to inhibit the phosphorylation of p38 which would cause the TLR2 and TLR4 receptors to be expressed less. This in turn would lead to the reduction of excessive inflammation of IBD.

In the study, the SRP-NPs were tested on in vitro cells and in vivo cells. Both of the groups showed a decrease in the SNX10 mRNA expression. For the in vitro cells, there was a decrease of 48% for the low dose and 95% for the high dose compared to the control group [3]. For the in vivo cells, there was a decrease of 52% for the low dose and 97% for the high dose compared to the control group. They also tested the SRP-NPs in two different mice group models, one with acute IBD and the other with chronic IBD. For both mice models, they found that for both of the groups, the SNX10 mRNA expression was found to be reduced when the mice models were given the nanoparticles [3]. Both of these tests suggested that the use of SNX10-shRNA in PLGA nanoparticles could be an effective treatment for IBD [3].

PLGA with Garcinol

Garcinol (GAR) is a natural compound and has been claimed to be a strong anti-inflammatory agent. The 2021 study done by Jacob et al. tested GAR loaded into a pH sensitive nanoparticle made up of PLGA that is coated with ES100 [12]. The particle created using the method of solvent evaporation was called GAR-PLGA-ES100. The purpose of the experiment was to test this particle in different pH conditions and test to see if the inflammatory effects were reduced.

To see how much the drug was released in vitro, the following equation was used [12]:

$$\% \text{ Release of GAR} = \frac{\text{Release of GAR}}{\text{Total GAR encapsulated inside NPs}} \times 100\%$$

The nanoparticles were tested at pH 1.2, 5.6 and 7.4. At pH 1.2 and 5.6 not a lot of drug was released with the ES100 coated particles compared to the non coated particles. It was found that at pH 7.4, there was the most release % of the GAR-PLGA-ES100 where after 48 hours in the pH conditions, 70.2 +/- 0.433% was released [12]. It is important to note that pH of 7.4 is approximately the pH of the colon. It was also shown that the nanoparticles were indeed uptaken by cells through the use of confocal microscopy and can be seen in Figure 7.

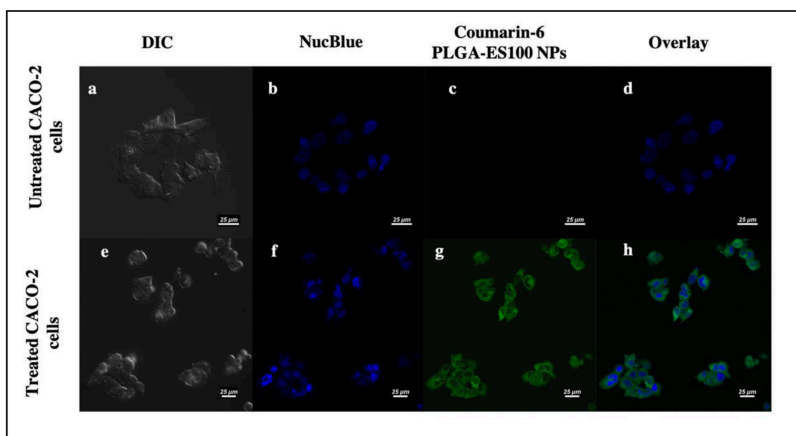


Figure 7: The cellular uptake of Coumarin-6 PLGA-ES100 nanoparticles [12].

It was also found that the ideal concentration of GAR in the GAR-PLGA-ES100 nanoparticles is between 50-100 micrograms/mL of GAR-PLGA-ES100 since at these concentrations, they exhibited the least cytotoxicity. Finally, both the MPO and LDH activity which was used to measure the inflammatory response where a higher activity present indicates it is more inflamed [12]. For the LDH activity, the use of 100 microgram/mL decreased LDH activity from 86.96% +/- 3.68 (the activity of the blank) to 40.04% +/- 8.8 and 250 microgram/mL reduced the LDH activity to 15.99% +/- 4.78 [12]. The free GAR reduced the LDH activity the most to 1.04% +/- 2.74 [12]. However, even though the free GAR reduced the activity the most for the LDH activity, for the MPO activity the free GAR was not as effective as the the GAR-PLGA-ES100 [12].

Implications

As observed above, the drug-loaded PLGA nanoparticles did not exhibit negative effects on the models' viability. Davoudi et al. (2017) reported that the organoids have shown normal growth in the presence or absence of PLGA nanoparticles. Ali et. al (2016) also confirmed from their findings that PLGA, or nanoparticles in general, have shown a high potential for targeted drug delivery in approaching IBD treatment. Should it be accompanied by a more advanced technology, the PLGA nanoparticle drug delivery system will achieve better rectal delivery properties [7]. Immunohistochemistry will be beneficial in determining the functionality and effectiveness of the drug delivery system [7].

In Li et al.'s study (2022), intravenous administration reduces the accumulative property of the nanoparticle as it tends to bind to S100A9 in the blood instead [14]. Oral administration allows direct delivery of the drug to the inflamed area, and this reduces any side reactions and increases the drug's concentrations among the local tissues, improving the anti-inflammatory response.

In the study conducted by Bao et al. (2021) had tested the use of shRNA plasmids which would directly interfere with the development of the inflammatory process, blocking TLR2 and TLR4 expression. Using this method, they comment on the results being promising, though there could be complications involved if the nanoparticle does not reach the colon [3].

In the study conducted by Jacob et al. (2021), they had tested a natural compound rather than the use of a synthesized drug or plasmids. They found that the PLGA particle should be coated with ES100 in order to survive the different pH levels of the GI tract. By doing so, the inflammation was able to be drastically reduced over a 48 hour time period.

In general, there are also some overarching limitations during the development of the nanoparticles including the drug efficiency and loading is not 100% perfect as well as the maximum amount of drug that is able to be loaded given the synthesization methods of the nanoparticles. Despite this, all of these studies suggest the promising treatment of IBD and also demonstrate the wide variety and options that patients are able to consider.

Conclusion

Inflammatory bowel disease (IBD) is a disease with no permanent cure and its symptoms affect patients' lives to an immense level, and at times, could even be life-threatening. The health risks and complications of IBD signify that there is a need to design and produce various drugs and drug delivery systems to treat it, or at least alleviate its symptoms. Studies show that using poly-co-glycolic acid (PLGA) for treating inflammatory bowel disease (IBD) exhibited promising results. Looking at PLGAs loaded with IBD treatment drugs and anti-inflammatory compounds, current literature specifically outlines that organoid models used have exhibited normal growth, or have no overall negative effect, with and without the presence of the nanoparticles.

Although there are promising developments on studies pertaining to IBD treatment using drug-loaded PLGA, there is not enough data to determine the most effective method of administration for the treatment. Reports have so far indicated that loading nanoparticles with drugs to be delivered to the lumen of the intestinal organoid demonstrates qualities that the drug delivery method is indeed beneficial in IBD treatment. Likewise, pharmaceutical researchers are looking forward to improving the technique or design of this delivery system for it to be tested clinically on IBD patients. Some reports have also mentioned that there are some limitations when preparing the nanoparticles such as there being a limit to how much of the drug can be loaded in the nanoparticles. Another facet to consider is the possible long-term effects of this treatment on the patient's body—whether the PLGAs accumulative tendency towards the target site have damaging effects to the organ, its cells, or neighboring environment. This is still an important factor that requires further research.

References

1. Abegunde, A. T., Muhammad, B. H., Bhatti, O., & Ali, T. (2016). Environmental risk factors for inflammatory bowel diseases: Evidence based literature review. *World Journal of Gastroenterology*, 22(27), 6296–6317. <https://doi.org/10.3748/wjg.v22.i27.6296>
2. Ali, H., Weigmann, B., Collnot, E. M., Khan, S. A., Windbergs, M., & Lehr, C. M. (2016). Budesonide loaded PLGA nanoparticles for targeting the inflamed intestinal mucosa—Pharmaceutical characterization and fluorescence imaging. *Pharmaceutical Research*, 33(5), 1085–1092. <https://doi.org/10.1007/s11095-015-1852-6>
3. Bao, W. L., Wu, Q., Hu, B., Sun, D., Zhao, S., Shen, X., Cheng, H., & Shen, W. (2021). Oral nanoparticles of SNX10-shRNA plasmids ameliorate mouse colitis. *International Journal of Nanomedicine*, 16, 345–357. <https://doi.org/10.2147/IJN.S286392>
4. Bhatia, S. (2016). Nanoparticles Types, Classification, Characterization, Fabrication Methods and Drug Delivery Applications. In: *Natural Polymer Drug Delivery Systems*. Springer, Cham. https://doi.org/10.1007/978-3-319-41129-3_2
5. Bilati, U., Allemann, E., & Doelker, E. (2005). Nanoprecipitation versus emulsion-based techniques for the encapsulation of proteins into biodegradable nanoparticles and process-related stability issues. *AAPS PharmSciTech*, 6(4), 594–604. <https://doi.org/10.1208/pt060474>
6. Chaudhary, S.A., Patel, D.M., Patel, J.K., Patel, D.H. (2021). Solvent Emulsification Evaporation and Solvent Emulsification Diffusion Techniques for Nanoparticles. In: Patel, J.K., Pathak, Y.V. (eds) *Emerging Technologies for Nanoparticle Manufacturing*. Springer, Cham. https://doi.org/10.1007/978-3-030-50703-9_12
7. Davoudi, Z., Peroutka-Bigus, N., Bellaire, B., Wannemuehler, M., Barrett, T. A., Narasimhan, B., & Wang, Q. (2017). Intestinal organoids containing poly(lactic-co-glycolic acid) nanoparticles for the treatment of inflammatory bowel diseases. *Journal of Biomedical Materials Research, Part A*, 106(4), 876–886. <https://doi.org/10.1002/jbm.a.36305>
8. De Jong, W. H., & Borm, P. J. (2008). Drug delivery and nanoparticles: applications and hazards. *International Journal of Nanomedicine*, 3(2), 133–149. <https://doi.org/10.2147/ijn.s596>
9. Dobrovolskaia, Marina A., & McNeil, Scott E. (2013). Handbook of Immunological Properties of Engineered Nanomaterials - 16.6 Types of Nanocarriers. (pp. 483-501). World Scientific. Retrieved from <https://app.knovel.com/hotlink/pdf/id:kt00C253EN/handbook-immunological/types-nanocarriers>

10. Ediriwickrema, A., & Saltzman, W. M. (2015). Nanotherapy for cancer: Targeting and multifunctionality in the future of cancer therapies. *ACS Biomaterials Science & Engineering*, 1(2), 64–78. <https://doi.org/10.1021/ab500084g>
11. Gentile, P., Chiono, V., Carmagnola, I., & Hatton, P. (2014). An overview of poly(lactic-co-glycolic) acid (PLGA)-based biomaterials for bone tissue engineering. *International Journal of Molecular Sciences*, 15(3), 3640–3659. <http://dx.doi.org/10.3390/ijms15033640>
12. Jacob, E. M., Borah, A., Pillai, S. C., & Kumar, D. S. (2021). Garcinol Encapsulated pH-sensitive biodegradable nanoparticles: A novel therapeutic strategy for the treatment of inflammatory bowel disease. *Polymers*, 13(6), 862. <https://doi.org/10.3390/polym13060862>
13. Leonard, S., Srinivasan, X., Liu, E.M., Collnot, M., Ferrari, C.M., & Lehr, B. Godin. (2020). Design and in vitro characterization of multistage silicon-PLGA budesonide particles for inflammatory bowel disease. *European Journal of Pharmaceutics and Biopharmaceutics*, 151, 61–72. <https://doi.org/10.1016/J.EJPB.2020.03.020>
14. Li, Z., Zhang, X., Liu, C., Peng, Q., Wu, Y., Wen, Y., Zheng, R., Yan, Q., & Ma, J. (2022). Macrophage-Biomimetic Nanoparticles Ameliorate Ulcerative Colitis through Reducing Inflammatory Factors Expression. *Journal of Innate Immunity*, 14(4), 380–392. <https://doi.org/10.1159/000519363>
15. Lü, J. M., Wang, X., Marin-Muller, C., Wang, H., Lin, P. H., Yao, Q., & Chen, C. (2009). Current advances in research and clinical applications of PLGA-based nanotechnology. *Expert Review of Molecular Diagnostics*, 9(4), 325–341. <https://doi.org/10.1586/erm.09.15>
16. Makadia, H. K., & Siegel, S. J. (2011). Poly Lactic-co-Glycolic Acid (PLGA) as Biodegradable controlled drug delivery carrier. *Polymers*, 3(3), 1377–1397. <https://doi.org/10.3390/polym3031377>
17. McCall, R. L., & Sirianni, R. W. (2013). PLGA nanoparticles formed by single- or double-emulsion with vitamin E-TPGS. *Journal of Visualized Experiments*, (82), 51015. <https://doi.org/10.3791/51015>
18. Mir, M., Ahmed, N., & ur Rehman, A. (2017). Recent applications of PLGA based nanostructures in drug delivery. *Colloids and Surfaces B: Biointerfaces*, 159, 217-231. <https://doi.org/10.1016/j.colsurfb.2017.07.038>
19. Naito, M., Yokoyama, T., Hosokawa, K., & Nogi, K. (2018). Nanoparticle Technology Handbook (3rd Edition) - 17.4.3.1 Drug-Eluting Stents. Elsevier. Retrieved from <https://app.knovel.com/hotlink/pdf/id:kt011PNP72/nanoparticle-technology/drug-eluting-stents>

20. National Center for Chronic Disease Prevention and Health Promotion. (2022). *What is inflammatory bowel disease (IBD)?*. Centers for Disease Control and Prevention. <https://www.cdc.gov/ibd/what-is-IBD.htm>
21. O'Donnell, S., & O'Morain, C. A. (2010). Therapeutic benefits of budesonide in gastroenterology. *Therapeutic Advances in Chronic Disease*, 1(4), 177–186. <https://doi.org/10.1177/2040622310379293>
22. Rousseaux, C., Lefebvre, B., Dubuquoy, L., Lefebvre, P., Romano, O., Auwerx, J., Metzger, D., Wahli, W., Desvergne, B., Naccari, G. C., Chavatte, P., Farce, A., Bulois, P., Cortot, A., Colombel, J. F., & Desreumaux, P. (2005). Intestinal antiinflammatory effect of 5-aminosalicylic acid is dependent on peroxisome proliferator-activated receptor-gamma. *The Journal of Experimental Medicine*, 201(8), 1205–1215. <https://doi.org/10.1084/jem.20041948>
23. Sagar R. Pardeshi, Aniket Nikam, Priyanka Chandak, Vijaya Mandale, Jitendra B. Naik & Prabhanjan S. Giram (2021): Recent advances in PLGA based nanocarriers for drug delivery system: a state of the art review, *International Journal of Polymeric Materials and Polymeric Biomaterials*, <https://doi.org/10.1080/00914037.2021.1985495>
24. Seyedian, S. S., Nokhostin, F., & Malamir, M. D. (2019). A review of the diagnosis, prevention, and treatment methods of inflammatory bowel disease. *Journal of Medicine and Life*, 12(2), 113–122. <https://doi.org/10.25122/jml-2018-0075>
25. Wooley, Dawn P., & Byers, Karen B. (2017). Biological Safety - Principles and Practices (5th Edition) - 12.5 Nanoparticles. (pp. 275-276). American Society for Microbiology (ASM). Retrieved from <https://app.knovel.com/hotlink/pdf/id:kt011EED18/biological-safety-principles/nanoparticles>
26. Yang, C., & Merlin, D. (2019). Nanoparticle-mediated drug delivery systems for the treatment of IBD: Current perspectives. *International Journal of Nanomedicine*, 14, 8875–8889. <https://doi.org/10.2147/IJN.S210315>
27. Yadav, K. S., & Sawant, K. K. (2010). Modified nanoprecipitation method for preparation of cytarabine-loaded PLGA nanoparticles. *AAPS PharmSciTech*, 11(3), 1456–1465. <https://doi.org/10.1208/s12249-010-9519-4>
28. Yamamoto-Furusho, J. K. (2007). Genetic factors associated with the development of inflammatory bowel disease. *World Journal of Gastroenterology*, 13(42), 5594–5597. <https://doi.org/10.3748/wjg.v13.i42.5594>
29. Zhang, Y. Z., & Li, Y. Y. (2014). Inflammatory bowel disease: Pathogenesis. *World Journal of Gastroenterology*, 20(1), 91–99. <https://doi.org/10.3748/wjg.v20.i1.91>